Kettering UNIVERSITY

College of Engineering and Computer Science
Department of Computer Science

CS 203 Computing and Algorithms III (Fall 2026)
Course Instructor: Dr. Jamal Alhiyafi

Group Activity # 1 (5%)
[Each group is 1-2 students]

Due: Thursday, 11/5/2026 @ 11:59 PM

Instructions: (In addition to code running correctly, points will be deducted if these instructions are not followed)

- Grade will be based on the running code. The code must be commented. A naming convention should be followed. Code must be readable, so make sure to use space, lines, indention, etc.
- Make sure to add your name(s), CS203, Fall 26, and Group Activity # 1 at the beginning of each file.
- Code files must refer to the description of the problem.
- For each program, you should create a text file called README.txt that contains information on how to compile and run the program.
- Put the code files and all their associated files, including the readme file, in a separate folder. The folder name should refer to the program description.
- Put both folders in a WinZip file or WinRAR file named:
  YourLastNames-CS203-F26-GroupActivity1-G#.winzip or .winrar.
- Submission should be via Blackboard.

(2.5 points) Write a Java program (based on a graph traversal algorithm you've learned in this class) that, for a given undirected graph, outputs: the vertices of each connected component.
Your programs should take inputs from a file via the command line (or by asking the user to provide the file name) with the following structure in the input file. Each line of the input file represents a graph. The first number in a line specifies the number of vertices in the graph. Then pairs of vertices define the edges.

An example of an input file is as follows:
5 (1, 2) (3, 4) (3, 5) (4, 5)
4 (1, 2) (2, 3) (1, 4)

It specifies two graphs. The first graph has five vertices (1, 2, 3, 4, 5) and four edges. The second graph has four vertices (1, 2, 3, 4) and three edges.

The proper output should look (something) like:

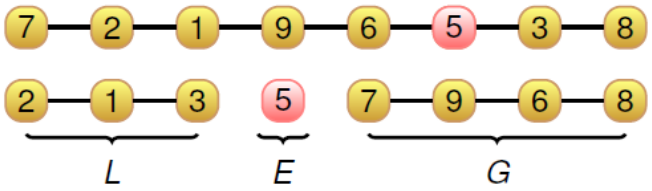
Graph1:
Two connected components: {1, 2} {3, 4, 5}

Graph2:
One connected component: {1, 2, 3, 4}

You must test your programs on a nontrivial input file (with at least 3 graphs and each graph having 7-10 nodes). Your output should be formatted nicely so that it is easy to read.

(2.5 points) Write a Java program that implements k-selection.

- The input is an unsorted array of length n.
- Find the value in the array that would be on position k if the array were sorted. You should not sort the array to find the position of k.
- For example, in the figure below, the number 5 is on position 4. Thus, for k=4, find the number that is on position 4; the result is 5.



- Submit the code that solves the problem and sample outputs that show the input array, the value of k, and the result (the value found).